Benchmarking Workflow Audit

Reliability Analysis of Literature Benchmarking for the SF_6 Global Model

AI–Researcher Task Boundaries and Manual Validation Requirements

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1 Benchmarking Workflow: Steps and Error Sources

Reproducing a published result requires a chain of steps. An error at any step propagates into the final comparison and can be misinterpreted as a model discrepancy.

1.1 Step-by-step reconstruction and error sources

#	Step	What is done	Where errors enter
1	Identify figures	Determine which figures to benchmark against (Figs. 5–8 in Kokkoris).	Misidentifying which panel contains the model curve vs experimental data.
2	Identify curves	For each figure, determine which line is the "basic" model, which is "no deposition," which is experiment.	Curves overlap. Colour on a printout differs from the PDF. Dashed vs solid indistinguishable at low resolution.
3	Read axis scales	Determine units, range, linear vs log scale, and dual-axis assignments.	Log axes: 0.5 cm on paper = 3× in density. Dual-axis: left vs right scale misassignment.
4	Extract data	Read numerical values from the curves (digitization).	**Highest error risk.** See Section 2.
5	Set operating conditions	Configure model to match paper: p_{OFF}, P, T_{gas}, flow rate, η.	Implicit assumptions in the paper (e.g. $\eta = 1$) that differ from the reconstruction.
6	Run model	Time-integrate to steady state.	Solver tolerances, convergence criteria, initial conditions.
7	Compare	Plot model vs digitized data on the same axes.	Comparing model output to incorrectly digitized data produces a "discrepancy" that is actually a data-extraction error.

Steps 1–4 are entirely about *reading the paper correctly.* Steps 5–6 are about *running the model correctly.* Step 7 combines both. The most dangerous errors are in steps 2–4 because they are silent: the comparison will look plausible even if the extracted data is wrong by a factor of 2.

2 Critique of the Current Digitization

2.1 How the data was extracted

The benchmarking data in `paper_digitized.json` was extracted by AI from uploaded screenshots of the paper figures. No pixel-level digitization tool (e.g. WebPlotDigitizer) was used. The AI estimated values by visually inspecting the images and writing down approximate numbers.

2.2 Evidence of estimation rather than measurement

- **All values are round numbers.** The SF_6 density at 3500 W is recorded as 5×10^{19}, not 4.7×10^{19} or 5.3×10^{19}. This is the hallmark of visual estimation, not precise extraction.
- **Log-scale precision is fundamentally limited.** On a plot spanning 10^{17} to 10^{21}, one millimetre on paper corresponds to approximately 0.1 decades, or a factor of 1.26× in density. Without pixel-level measurement, the best achievable accuracy is ±0.2 decades (±60%).
- **No validation overlay was produced.** The extracted points were never plotted back on top of the original figure to verify that they lie on the correct curve. This is the single most important missing step.

2.3 Specific high-risk figures

Figure	Risk level	Issue
Fig. 7a	Moderate	6 curves, but well-separated and colour-coded. SF_6 and F cross at high power — if AI reads the wrong curve after the crossing, errors of 3× are possible.
Fig. 7b	**HIGH**	6 curves on 1 decade of log scale. SF_5^+ (blue), F^- (green), SF_3^+ (magenta), SF_6^- (red), SF_4^+ (uncertain), e^- (brown). Curves nearly overlap in the 10^{16}–10^{17} range. Misidentifying one curve for another gives a factor of 2–5× error.
Fig. 7c	**HIGH**	Dual y-axis. Left: α (0–45, linear). Right: T_e (3.0–5.5 eV). *Two* α curves (Druyvesteyn and Maxwellian) and *two* T_e curves. Reading the Maxwellian T_e instead of the Druyvesteyn T_e gives a 0.5–0.8 eV error.
Fig. 8a	Moderate	Same as 7a but with different species ordering at high p_{OFF}.
Fig. 8b	**HIGH**	Same issue as 7b but curves are even more compressed.
Fig. 8c	**HIGH**	Same dual-axis, four-curve issue as 7c. α at 4.5 Pa: caption says ∼25, digitized as 30.

2.4 How digitization errors can create false discrepancies

Consider the F density in the pressure sweep (Fig. 8a). The digitized paper curve shows F increasing from 6×10^{19} at 0.1 Pa to 1.08×10^{20} at 2.5 Pa. Our model shows F decreasing from 8×10^{19} to 6.4×10^{19}. This was interpreted as a qualitative disagreement (opposite trends).

However, if the digitized F values are wrong by even 30% (well within the estimation error for log-scale plots), the actual paper F density might be 7–8×10^{19} at 2.5 Pa rather than 1.08×10^{20}. The "opposite trend" discrepancy could partially be a digitization artifact amplified by a 3-decade log scale.

This does not mean the discrepancy is imaginary — the physics analysis in the discrepancy report is sound regardless. But the *magnitude* of the discrepancy, and hence the conclusions about which physics to fix, could be misleading.

3 Redesigned Benchmarking Workflow

The following workflow replaces the ad hoc approach used previously with a validated pipeline.

3.1 Phase 1: Figure identification (AI + researcher)

1. AI identifies all relevant figures in the paper and lists: figure number, panel letter, quantity plotted, independent variable, axis type (linear/log), units, number of curves, and operating conditions.
2. **Researcher reviews this list** and confirms each entry by inspecting the paper. Corrects any misidentifications (e.g. AI calling a Maxwellian curve "Druyvesteyn").

3.2 Phase 2: Curve extraction (researcher only)

3. Researcher uses **WebPlotDigitizer** or an equivalent calibrated tool to extract data points from each curve. The tool requires manual calibration of the axes (selecting two known points on each axis) and manual identification of which curve to trace.
4. For each extracted dataset, the researcher exports a CSV file with columns: x (independent variable), y (extracted value), and a note identifying which curve it came from.
5. **Validation overlay:** the researcher plots the extracted points on top of the original figure image (using WebPlotDigitizer's overlay feature or a Python script) and visually confirms that every point sits on the intended curve. Any point that appears to be on the wrong curve is re-extracted.

3.3 Phase 3: Condition matching (AI + researcher)

6. AI configures the model to run at the same operating conditions as each figure. AI lists the conditions used and the researcher confirms they match the paper.
7. **Researcher checks:** correct pressure (p_{OFF} vs p_{total}), correct power definition (source power vs absorbed power), correct gas temperature, and correct flow rate.

3.4 Phase 4: Comparison (AI + researcher)

8. AI generates overlay plots with model and extracted data on the same axes.
9. AI reports quantitative comparisons (ratios, offsets).
10. **Researcher reviews** each overlay and confirms: (a) the extracted data looks correct, (b) the model output is labelled correctly, (c) the axis scales match, (d) the comparison is between the correct quantities.

4 AI Tasks vs Researcher Tasks

4.1 Tasks AI can perform reliably

Task	Why AI is reliable
Paper reading and equation extraction	Text is unambiguous. Equations can be cross-checked against dimensional analysis.
Reaction mechanism reconstruction	Tabulated data (Table 1) is structured and can be transcribed exactly.
Rate coefficient implementation	Functional form (Eq. 1) is explicit. Parameters are in a table.
Code generation and debugging	Deterministic: code either runs and gives the correct output or it doesn't.
Parameter sweeps and sensitivity analysis	Automated: AI runs the model at many conditions and tabulates results.
Figure generation	Plotting from CSV data is deterministic.
Trend identification and physical interpretation	AI can correctly identify monotonic, saturating, or non-monotonic behaviour and explain the physics.
LaTeX report generation	Formatting and typesetting.

4.2 Tasks that must be performed by the researcher

Task	Why AI cannot do this reliably
Curve identification in multi-curve plots	AI cannot reliably distinguish 6 overlapping curves on a 1-decade log plot from a low-resolution screenshot. Colour perception, line style identification, and label-to-curve association are unreliable.
Axis calibration for data extraction	Requires clicking on known axis points with pixel precision. AI has no pixel-level control over images.
Validation overlay inspection	Requires side-by-side comparison of extracted points on the original figure at full resolution. AI can generate the overlay, but only a human can judge whether each point is on the correct curve.
Confirming experimental vs model curves	Papers often show both on the same plot with similar styles. Only the researcher (who understands the context) can reliably distinguish them.
Operating condition verification	Requires checking whether p_{OFF} in the paper means the same as in the model. This involves domain knowledge that AI may not correctly resolve from ambiguous text.
Dual-axis reading	On Fig. 7c, both α (left) and T_e (right) share the same x-axis. There are 4 curves. AI cannot reliably determine which curve corresponds to which axis.
Cross-checking against figure captions	The caption for Fig. 8c states "ratio increases from ∼0.03 at 0.1 Pa to 25 at 4.5 Pa." The digitized value at 4.5 Pa is 30, not 25. Only a careful human reading catches this.

4.3 The critical boundary

The boundary is simple: **AI generates, the researcher validates.** AI can produce a candidate digitization, a candidate model run, and a candidate comparison plot. But before any scientific conclusion is drawn, the researcher must manually confirm that the data being compared is correct. This is analogous to the requirement in experimental science that raw data must be inspected by the experimentalist before statistical analysis is performed.

5 Manual Validation Checklist

The following checklist should be completed by the researcher before accepting any benchmarking result. Each item should be checked off with initials and date.

5.1 For each extracted curve

	Validation item
1	I have identified which curve in the figure corresponds to the quantity I am extracting (e.g. "Druyvesteyn T_e," not "Maxwellian T_e").
2	I have confirmed the axis type: linear or logarithmic. For log axes, I have verified the decade markings.
3	I have confirmed the axis units (m^{-3}, Pa, eV, W).
4	For dual-axis plots, I have confirmed which curve corresponds to the left axis and which to the right axis.
5	I have used a calibrated digitization tool (not visual estimation) to extract the data points.
6	I have plotted the extracted points on top of the original figure and confirmed that each point lies on the intended curve.
7	I have compared the extracted endpoint values against any numerical values stated in the figure caption or paper text.

5.2 For each model comparison

	Validation item
8	The model operating conditions match the paper: power (source or absorbed), pressure (p_{OFF} or p_{total}), gas temperature, flow rate.
9	The model output quantity matches the paper's plotted quantity (e.g. number density, not partial pressure; total negative ion density, not individual species).
10	The x-axis variable is the same (e.g. p_{OFF}, not total pressure; source power, not absorbed power).
11	The model has reached steady state (checked by verifying that all derivatives are below a convergence threshold).
12	The comparison is against the paper's *model* curve, not the experimental data, unless the comparison is explicitly intended to be against experiment.

5.3 Before drawing conclusions

	Validation item
13	Any discrepancy larger than 2× has been checked for digitization error before being attributed to model physics.
14	For log-scale comparisons, I have verified that a reported "3× discrepancy" is not within the ±0.5-decade estimation error of the extracted data.
15	The paper's own experimental scatter (visible as spread in data points) has been compared with the model–paper discrepancy. If the discrepancy is smaller than the experimental scatter, it should not be over-interpreted.

6 Specific Audit of the Current SF_6 Benchmarking

Applying the checklist above to the current digitization reveals the following issues.

Checklist item	Status	Risk	Action needed
1: Curve identification	Probable	Moderate	Re-examine Fig. 7b, 8b for curve–species mapping.
2: Axis type	**Done**	Low	Log axes correctly identified.
3: Axis units	**Done**	Low	m^{-3}, Pa, eV all correct.
4: Dual-axis	Uncertain	**High**	Fig. 7c, 8c: Druyvesteyn vs Maxwellian not confirmed.
5: Calibrated tool	**Not done**	**High**	Must re-extract with WebPlotDigitizer.
6: Validation overlay	**Not done**	**High**	Must produce and inspect overlays.
7: Caption cross-check	Partial	Moderate	α at 4.5 Pa: caption says 25, digitized as 30.
8–10: Conditions	**Done**	Low	Conditions match.
11: Steady state	**Done**	Low	BDF converges.
12: Model vs experiment	**Done**	Low	Comparison is against model curve.
13–14: Discrepancy check	**Not done**	**High**	Large discrepancies not verified against extraction error.

The four **HIGH** items (5, 6, 13–14) mean that the current benchmarking conclusions should be treated as **provisional** until the researcher re-extracts the data with a calibrated tool and produces validation overlays.

6.1 Recommended immediate actions

1. **Re-extract Fig. 7c and 8c using WebPlotDigitizer.** These dual-axis plots are the source of α and T_e comparison data. If T_e was read from the Maxwellian curve instead of the Druyvesteyn curve, the entire T_e "discrepancy" of 0.15 eV could be a digitization artifact.
2. **Re-extract Fig. 7b and 8b.** These charged-species plots have the most curve overlap. Verify SF_5^+ and F^- identification.
3. **Produce validation overlays for all extracted datasets.** Plot the extracted CSV points on top of the original figure images and include these overlays in the benchmarking report.
4. **Cross-check all extracted values against the paper text.** The paper states specific numbers in the abstract, results section, and figure captions. Use these as calibration anchors.

7 References

1. Kokkoris et al., *J. Phys. D: Appl. Phys.* **42**, 055209 (2009)
2. Rohatgi, A., *WebPlotDigitizer*, Version 4.7 (2024), https://automeris.io/WebPlotDigitizer